

AI Technical Reference

How the AI works · body-point detection · measurement maths · accuracy & the on-screen visual layers

This document explains the computer-vision pipeline behind the HealingMonk assessment: the neural pose model used, how the 33 body landmarks are located and marked on the patient, how joint angles and the clinical plumb line are computed, what the accuracy limits are, and what every line and colour on the live overlay means. It is intended for clinicians, reviewers and engineers.

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1 · The AI at a glance

HealingMonk turns an ordinary camera into a clinical-style movement lab. Every measurement runs entirely on the patient's device (in the browser) — no video is uploaded. The engine detects the body, tracks 33 points in 3D in real time, and converts their geometry into clinical measurements (angles, tilts, range-of-motion and a plumb-line alignment score).

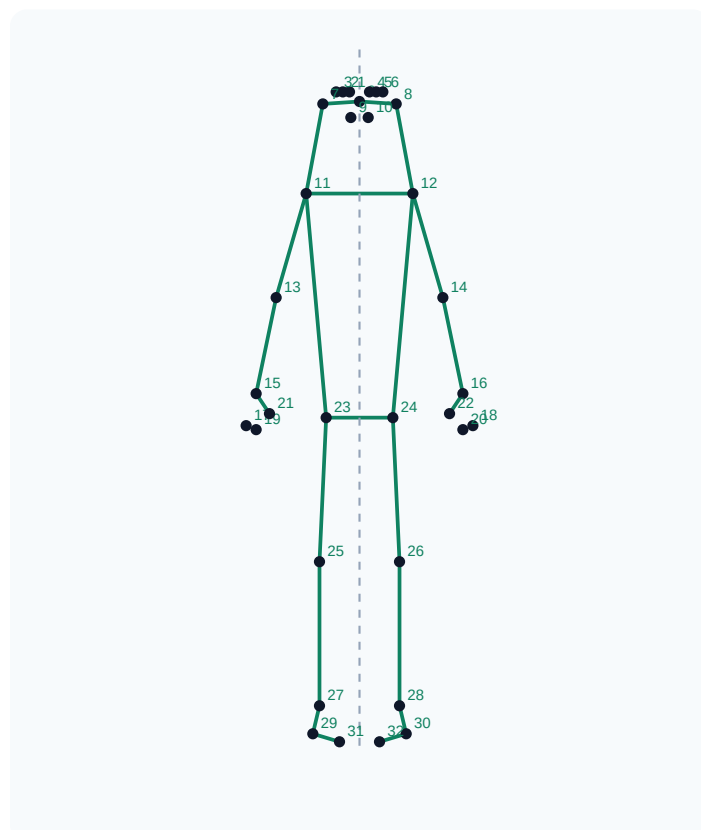
Model	Google MediaPipe Pose Landmarker — "full" variant (float16)
Runtime	@mediapipe/tasks-vision 0.10.35 (WebAssembly + WebGL)
Landmarks	33 full-body points, each with x, y, z (depth) and visibility
Where it runs	Client-side in the browser — camera frames never leave the device
Acceleration	GPU delegate when available, automatic CPU fallback
Mode	VIDEO / real-time, single person (numPoses = 1)
Detection thresholds	pose-detection, presence & tracking confidence all "e 0.60
Smoothing	One-Euro adaptive filter on every point (x, y, z)
Clinical library	34 assessments across posture, upper limb, trunk, lower limb, seated & lying

Important: the automatic numbers are camera-based estimates, not a medical diagnosis. The treating doctor reviews every finding and can override the score — that clinical judgement is the record.

2 · How the body points are detected & marked

The pose model is a two-stage convolutional neural network. A detector first finds the person in the frame; a landmark network then regresses the pixel location of 33 predefined anatomical points, plus a depth value (z) and a visibility confidence (0-1) for each. HealingMonk uses the "full" model, which tracks limbs and depth markedly better than the "lite" model while still running in real time.

The 33 landmarks



- 0-10 Face / head
- 11-22 Shoulders, arms, hands
- 23-28 Hips, knees, ankles
- 29-32 Heels & toes
- - - Plumb reference line

Each of the 33 points carries x, y, z (depth) and a 0-1 visibility score. Points below 0.5 visibility are ignored so a hidden limb never corrupts a measurement.

Every point is returned in normalized coordinates (0-1 across the frame), so the maths is resolution-independent. On the live view each point is drawn as a green dot and the joints are joined into a skeleton; the same overlay is "baked into" the captured photo so the doctor sees exactly what the model measured.

All 33 points — where each is marked & what it measures

The complete list below maps every landmark index to its name, the body part it is marked on, and how HealingMonk uses it. Use this as the quick reference when explaining the system.

#	Point name	Body part	Marked on / used for
0	nose	Head	Head centre — forward-head check & side-view facing direction
1	left_eye_inner	Head	Eye landmark — helps estimate vertex (top of head)
2	left_eye	Head	Left eye — head tilt & vertex estimate
3	left_eye_outer	Head	Outer eye corner
4	right_eye_inner	Head	Eye landmark
5	right_eye	Head	Right eye — head tilt & vertex estimate
6	right_eye_outer	Head	Outer eye corner
7	left_ear	Head	Left ear (EAM) — craniovertebral angle & side plumb ear point
8	right_ear	Head	Right ear (EAM) — craniovertebral angle & side plumb
9	mouth_left	Head	Mouth corner — used to estimate the chin point
10	mouth_right	Head	Mouth corner — used to estimate the chin point
11	left_shoulder	Shoulder / torso	Acromion — shoulder level, plumb line, arm-ROM origin
12	right_shoulder	Shoulder / torso	Acromion — shoulder level, plumb line, arm-ROM origin
13	left_elbow	Arm	Elbow — elbow flexion & shoulder ROM vertex
14	right_elbow	Arm	Elbow — elbow flexion & shoulder ROM vertex
15	left_wrist	Arm	Wrist — upper-limb ROM endpoint
16	right_wrist	Arm	Wrist — upper-limb ROM endpoint
17	left_pinky	Hand	Little-finger knuckle
18	right_pinky	Hand	Little-finger knuckle
19	left_index	Hand	Index finger — reach / hand endpoint
20	right_index	Hand	Index finger — reach / hand endpoint
21	left_thumb	Hand	Thumb
22	right_thumb	Hand	Thumb
23	left_hip	Pelvis	Hip / ASIS — pelvic tilt, trunk & hip ROM, plumb line
24	right_hip	Pelvis	Hip / ASIS — pelvic tilt, trunk & hip ROM, plumb line
25	left_knee	Leg	Knee — knee alignment/flexion, squat depth, plumb line
26	right_knee	Leg	Knee — knee alignment/flexion, squat depth, plumb line
27	left_ankle	Leg	Ankle (lateral malleolus) — plumb base & dorsiflexion
28	right_ankle	Leg	Ankle (lateral malleolus) — plumb base & dorsiflexion
29	left_heel	Foot	Heel — foot alignment & ankle ROM
30	right_heel	Foot	Heel — foot alignment & ankle ROM
31	left_foot_index	Foot	Toe — foot direction & dorsiflexion
32	right_foot_index	Foot	Toe — foot direction & dorsiflexion

Visibility gate: any point whose visibility drops below 0.5 is treated as "not seen" and excluded, so a briefly hidden hand or foot can never pull a measurement to a wrong value. A live detection-quality indicator (visible points / 33) tells the patient when to reposition.

3 · Steadying the signal (why the points don't shimmer)

Raw neural landmarks jitter by a pixel or two every frame even when the subject is perfectly still, which would

make the skeleton shimmer and the angles flicker by $\pm 1-2^\circ$. HealingMonk filters every landmark axis with a One-Euro adaptive filter (Casiez et al., CHI 2012):

- When the point is slow / still, it smooths hard — removing all jitter so the deviation line sits rock-steady.
- When the point moves fast, it barely smooths — so there is no visible lag when the patient actually moves.

Tuned constants (per source): baseline cutoff 0.9 Hz, speed coefficient $^2 = 0.0001$. The pose of x, y and z is filtered independently; the filter resets when the body leaves the frame so a re-acquired pose never snaps.

4 · How the measurements are computed

Joint angles

Any joint angle is the angle at point B in the triangle A-B-C (e.g. shoulder-elbow-wrist), computed from the 3D vectors:

```
angle = arccos( (BA · BC) / (|BA| × |BC|) ) × 180 / Å
BA = A " B      BC = C " B
· = dot product  |v| = vector length
Result clamped to a valid 0-180° range.
```

A 2D variant (ignoring z) is used where the clinically correct plane is the image plane — e.g. shoulder level or head tilt. Craniovertebral angle, shoulder flexion/abduction, knee flexion, pelvic tilt and trunk ROM are all derived this way and update live as the patient moves.

Clinical plumb line

A plumb line is the vertical reference a physiotherapist drops through the body to screen standing posture.

HealingMonk reconstructs it per view (front / side / back) and reports how far each anatomical checkpoint sits off that line, as a percentage of body height — so the score is scale-free.

- Front: central midline through vertex !' nose !' chin !' sternum !' umbilicus !' p
plus left/right symmetry (shoulders, clavicle, ASIS, knees, feet).
- Side: ear (EAM) !' shoulder (acromion) !' hip (greater trochanter) !' knee !' an
- Back: occiput !' cervical/thoracolumbar spine !' gluteal cleft !' knee & ankle

MediaPipe only returns 33 surface points, so several true clinical landmarks (vertex, sternum, umbilicus, occiput, greater trochanter) are estimated geometrically from the points that exist. Tolerances: central/sagittal checkpoints within $\pm 4\%$ (front/back) or $\pm 5\%$ (side) of body height; left/right pairs level within 3° .

Severity rating

The plumb score (mean absolute checkpoint offset) is graded into the same four-band scale used across the report:

score	"d	2 %	!	Normal	(green)
score	"d	4 %	!	Mild	(yellow)
score	"d	7 %	!	Moderate	(orange)
score	>	7 %	!	Severe	(red)

Overall posture health score (report)

The headline 0-100 score on the report is a weighted deviation from ideal across all captured findings:

```
penalty(finding) = 0 normal . 1 mild . 2 moderate . 3 severe
score = 100 - ( £ penalty / ( 3 × findings ) ) × 100
clamped to 0-100 and rounded.
```

5 · Accuracy, feasibility & limitations

Across the 34-assessment library each posture is tagged with an AI-feasibility rating that reflects how reliably a single camera can measure it: most posture, upper-limb and trunk screens are rated High; a few depth- or rotation-sensitive measures (e.g. anterior pelvic tilt, spinal rotation) are rated Partial or Medium and are flagged for extra doctor scrutiny.

High feasibility	Movement is clearly visible in the camera plane — reliable estimate (majority of the library).
Medium	Usable, but sensitive to camera angle or subtle motion — verify against exam.
Partial	Depth / rotation limited by a single 2D camera — treat as a screening indicator only.

What affects accuracy in practice

- Lighting — natural or bright even light is best; low light or backlighting reduces landmark confidence.
- Framing & distance — the full body (or the relevant segment) must be in frame; a live quality bar guides repositioning.
- Clothing — contrasting, non-baggy clothing gives the cleanest landmarks.
- Depth ambiguity — a single camera estimates z (depth), so purely front-back movements are the hardest and are rated Partial.

Performance envelope: roughly 50-90 ms per frame end-to-end (detection + angles + rendering), with GPU acceleration when the device allows. Because everything is an estimate, the report carries an explicit AI disclaimer and the doctor's clinical score overrides the automatic value on every finding.

6 · The on-screen visuals (3D points, skeleton & animation)

Several layers are drawn live over the camera feed and preserved in the report so the assessment is fully transparent:

- 3D landmark dots — all visible points from the 33-point model, positioned using the model's x, y and z (depth) output.
- Skeleton overlay — green lines joining the points (shoulders ! elbows ! wrists ! ankles !) so the tracked body is obvious at a glance.
- Moving body axis — the patient's actual head ! foot centre line, drawn green, showing where it drifts off, giving a live "correct / off" read.
- Fixed plumb reference — a dashed vertical in the severity colour, with each anatomical checkpoint labelled by its deviation (e.g. "Ear 6% ant").
- Symmetry bars (front view) — dashed left ! right connectors showing should be high.
- Muscle-map illustration — for each posture the report shows an anatomical figure (anterior or posterior) with the targeted muscle groups highlighted, so the patient sees which area the assessment relates to.

Together these turn a raw camera frame into a readable clinical picture: the patient photo with the AI points baked in, next to the ideal reference — exactly what the doctor scores against.

Appendix · The full assessment library

Every posture, range-of-motion and movement screen the engine can run, with the single-camera measurement it produces and its AI-feasibility rating. Feasibility colour: green = High, purple = Medium, amber = Partial. (34 assessments.)

#	Assessment	Region	Measurement	Feasibility
1	Full Body — Center (Front)	Full Body	Overall Alignment (°)	High
2	Full Body — Back	Full Body	Overall Alignment (°)	High
3	Full Body — Left Side	Full Body	Trunk Vertical Lean (°)	High
4	Full Body — Right Side	Full Body	Trunk Vertical Lean (°)	High
5	Forward Head Posture	Cervical	Craniovertebral Angle (°)	High
6	Shoulder Level Symmetry	Shoulder	Shoulder Tilt Angle (°)	High
7	Pelvic Level Symmetry	Hip	Pelvic Tilt Angle (°)	High
8	Lateral Spinal Alignment	Thoracic	Trunk Lateral Deviation (°)	Partial
9	Lateral Head Tilt	Cervical	Head Tilt Angle (°)	High
10	Knee Alignment (Valgus/Varus)	Knee	Knee Deviation Angle (°)	High
11	Shoulder Flexion ROM (Overhead Reach)	Shoulder	Shoulder Flexion Angle (°)	High
12	Thoracic Kyphosis (Upper-Back Rounding)	Thoracic	Upper-Trunk Rounding (°)	Partial
13	Anterior Pelvic Tilt	Pelvis	Trunk–Thigh Tilt (°)	Partial
14	Side Plank (concave side)	Spine	Trunk Asymmetry (°)	Partial
15	Squat Depth (Knee Flexion)	Knee	Knee Flexion (°)	High
16	Hip Abduction ROM (Side Leg Raise)	Hip	Hip Abduction Angle (°)	High
17	Shoulder Abduction ROM (Side Arm Raise)	Shoulder	Shoulder Abduction Angle (°)	High
18	Elbow Flexion ROM	Elbow	Elbow Flexion (°)	High
19	Trunk Forward Flexion (Toe Touch)	Spine	Forward Bend Angle (°)	High
20	Trunk Lateral Flexion (Side Bend)	Spine	Side Bend Angle (°)	High
21	Cervical Flexion (Chin to Chest)	Cervical	Neck Flexion Angle (°)	Medium
22	Cervical Extension (Look Up)	Cervical	Neck Extension Angle (°)	Medium
23	Shoulder Extension ROM (Arm Back)	Shoulder	Shoulder Extension Angle (°)	High
24	Hip Flexion ROM (Knee Raise)	Hip	Hip Flexion Angle (°)	High
25	Active Knee Flexion (Heel to Buttock)	Knee	Knee Flexion (°)	High
26	Forward Lunge Depth	Knee	Front Knee Flexion (°)	High
27	Ankle Dorsiflexion (Knee to Wall)	Ankle	Ankle Angle (°)	Medium
28	Single-Leg Stance (Pelvic Drop)	Hip	Pelvic Drop (°)	High
29	Overhead Squat (Knee Control)	Knee	Knee Valgus Deviation (°)	High
30	Standing Back Extension	Spine	Back Extension Angle (°)	Medium

#	Assessment	Region	Measurement	Feasibility
31	Seated Forward Head (Sitting)	Cervical	Seated Craniovertebral Angle (°)	High
32	Seated Slump / Upper-Back Rounding (Sitting)	Thoracic	Seated Trunk Rounding (°)	Partial
33	Straight Leg Raise — SLR (Lying)	Hip	SLR Hip Flexion (°)	High
34	Knee-to-Chest Hip Flexion (Lying)	Hip	Hip Flexion (knee bent) (°)	High